TED (21) – 5083A REVISION 2021

MODEL QUESTION PAPER 1

S5 INSTRUMENTATION ENGINEERING

COURSE: INDUSTRIAL ELECTRONICS AND DRIVES

Time: 3 Hour *Max. Marks: 75*

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	Cognitive Level
1	Define the term latching current.	M1.02	U
2	The number of p-n junctions in a thyristor is ………………	M1.02	R
3	When UJT is used for triggering an SCR, the wave shape of the voltage obtained from UJT circuit is a a) sine wave b) sawtooth wave c) trapezoidal wave d) square wave	M2.01	U
4	Define Fleming's Left Hand rule.	M3.01	U
5	Write the back emf equation of a DC motor.	M3.01	U
6	Write the types of self-excited DC motor.	M3.02	R
7	A chopper, where voltage as well as current remain negative is known as type…………….. chopper	M4.02	R
8	Define a chopper.	M4.02	U
9	……..... is the stationary part of an induction motor	M3.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	Draw the two transistor model of an SCR.	M1.02	U
2.	Draw the static characteristics of DIAC.	M1.03	U
3.	Draw the circuit for RC triggering of an SCR.	M2.01	U
4.	Draw the circuit diagram of a single phase bridge inverter.	M2.03	U
5.	Write any three applications of stepper motor.	M3.03	U
6.	Write any three methods of speed control of induction motors.	M3.03	U
7.	What are the output control methods of a chopper?	M4.02	U
8.	Draw the circuit diagram of a single phase dual converter.	M4.01	U
9.	Write any three triggering methods of SCR.	M2.01	U
10.	Define the terms turn on time, turn off time and latching current of an SCR.	M1.02	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	*Cognitive Level*
III	Explain the VI characteristics of SCR.	M1.02	U
	OR		
IV	Describe the working of a TRIAC.	M1.03	U
V	With neat circuit diagram, explain R triggering of an SCR	M2.03	U
	OR		
VI	Describe TRIAC light dimming circuit.	M2.03	U
VII	Explain the working of AC tacho generators.	M3.03	U
	OR		
VIII	Explain the working of a stepper motor.	M3.04	U
IX	With circuit diagram and waveforms, step down chopper	M4.02	U
	OR		
X	Explain speed control of DC drives	M4.03	U
XI	Compare natural and forced commutation of SCR..	M2.03	U
	OR		
XII	Describe SCR triggering circuit using UJT.	M2.01	U
XIII	Explain Jones chopper..	M4.03	U
	OR		
XIV	Compare AC and DC drives.	M4.02	U

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S5 INSTRUMENTATION ENGINEERING

COURSE: INDUSTRIAL INSTRUMENTATION

Time: 3 Hour *Max. Marks: 75*

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	Cognitive Level
1	Define the term holding current.	M1.01	R
2	A TRIAC is equivalent to …………..SCR'sconnected in parallel.	M1.03	R
3	……….converts dc power into ac power at desired output voltage and frequency.	M2.03	U
4	A switch having no moving parts is called………..	M2.03	R
5	Class A or load commutation is possible in case of a) DC circuits only b) AC circuits only c) Both DC and AC circuits d) None of these	M2.02	U
6	…….... is the stationary part of an induction motor	M3.03	R
7	………… device is used to obtain variable DC voltage from a constant DC voltage source.	M4.02	U
8	Name the device which converts AC power of fixed frequency to AC power of different frequency	M4.01	R
9	………………. is used to prevent unwanted dv/dt triggering of SCR.	M2.02	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	Define a thyristor and sketch its V-I characteristics.	M1.02	U
2.	Define the terms turn on time, turn off time and latching current of an SCR.	M1.02	U
3.	Draw the circuit for R triggering of an SCR.	M2.01	U
4.	Draw the TRIAC light dimming circuit.	M2.03	U
5.	Write any three applications of single phase induction motor.	M3.03	U
6.	Write the equations for back emf and torque of a DC motor.	M3.01	U
7.	Write any three applications of chopper	M4.02	U
8.	Draw the circuit diagram of a three phase dual converter.	M4.01	U

9.	Draw the circuit diagram and waveforms for a single phase ac switch with R load	M2.04	U
10.	Draw the VI characteristics of a power diode.	M1.01	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	*Cognitive Level*
III	With a neat sketch, describe the two transistor analogy of an SCR.	M1.02	U
	OR		
IV	Describe the structure and working of a DIAC.	M1.03	U
V	With neat circuit diagram, explain RC triggering of an SCR	M2.03	U
	OR		
VI	With neat circuit diagram, explain triggering of an SCR using UJT	M2.03	U
VII	Explain the working of a single phase induction motor	M3.03	U
	OR		
VIII	Explain the working of a field controlled DC servo motor.	M3.04	U
IX	With circuit diagram and waveforms, explain Jone’s chopper	M4.02	U
	OR		
X	Explain speed control of DC drives	M4.03	U
XI	Compare AC and DC drives	M4.03	U
	OR		
XII	With circuit diagram and waveforms, explain single phase dual converter	M4.01	U
XIII	Explain principle and operation of a DC tachogenerators.	M3.04	U
	OR		
XIV	Explain the working of Universal motor.	M3.03	U